

Lesson 01: Square and Cube Roots

Name: _____

Class: MYP 8A

Date: _____

Learning objective

I can **calculate** and **estimate** square roots and cube roots.

Success criteria: I can ...

SC1 find the square root of a perfect square and the cube root of a perfect cube;**SC2** find the cube root of a negative number and explain why a negative number has no real square root;**SC3** say which two whole numbers a root lies between, and why;**SC4** estimate a root to 1 decimal place and check by squaring or cubing.

Key vocabulary

square root $\sqrt{\quad}$	$\sqrt{n}$ is the number that is <i>not negative</i> and multiplied by itself gives n . $\sqrt{49} = 7$ because $7^2 = 49$.
cube root $\sqrt[3]{\quad}$	$\sqrt[3]{n}$ is the number that, cubed, gives n . $\sqrt[3]{8} = 2$ because $2^3 = 8$.
radical sign	The symbol $\sqrt{\quad}$. A small 3 means cube root: $\sqrt[3]{\quad}$.
perfect square / cube	A whole number squared / cubed, e.g. $36 = 6^2$, $125 = 5^3$.
inverse operation	An operation that undoes another. Square root undoes squaring; cube root undoes cubing.
estimate	A sensible approximate value. We write $\approx$ ("is approximately equal to").

Section 1: Starter

independent · 3 minutes

S1. $7^2 = \underline{\hspace{2cm}}$

S2. $11^2 = \underline{\hspace{2cm}}$

S3. $3^3 = \underline{\hspace{2cm}}$

S4. $10^3 = \underline{\hspace{2cm}}$

S5. $(-4)^2 = \underline{\hspace{2cm}}$

S6. Which number is both a perfect square and a perfect cube: 16, 27, 64, 100? $\underline{\hspace{2cm}}$

Section 2: Notice and Why

*together*N1. Each shape is a square. The number inside is its area in cm^2 .

side = 4 cm



side = ?



side = 5 cm

(a) Finding the side from the area *undoes* squaring. Complete: $\sqrt{16} = \underline{\hspace{1cm}}$ and $\sqrt{25} = \underline{\hspace{1cm}}$ (b) The side of the middle square is between $\underline{\hspace{1cm}}$ cm and $\underline{\hspace{1cm}}$ cm. Is it closer to 4 or to 5? Why?

Section 3: I DO: worked examples

watch, then annotate

Example 1: Calculate roots of perfect squares and cubes

Working

Reason / check

(a) $\sqrt{81} = 9$

Ask: which number times itself is 81? Check: $9^2 = 81$ ✓

(b) $\sqrt[3]{125} = 5$

Ask: which number cubed is 125? Check: $5 \times 5 \times 5 = 125$ ✓

(c) $\sqrt[3]{-8} = -2$

A negative number cubed is negative. Check: $(-2)^3 = -8$ ✓

(d) $\sqrt{-9}$: no real answer

 $3^2 = 9$ and $(-3)^2 = 9$. Any real number squared is 0 or positive, so nothing squares to -9 .

My notes:

Example 2: Estimate $\sqrt{50}$ to 1 decimal place

Working

Reason

1. **Trap:** $49 < 50 < 64$

Find the perfect squares just below and just above 50.

2. **Roots:** $7 < \sqrt{50} < 8$

Take the square root of each: $\sqrt{49} = 7$, $\sqrt{64} = 8$.

3. **Closer?** closer to 49

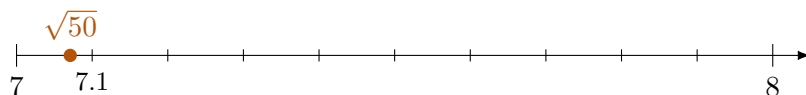
50 is 1 away from 49 but 14 away from 64, so $\sqrt{50}$ is just over 7.

4. **Test:** $7.1^2 = 50.41$

 $7.0^2 = 49$ is below 50, and $7.1^2 = 50.41$ is above 50, so $\sqrt{50}$ is between 7.0 and 7.1.

5. **Decide:** $\sqrt{50} \approx 7.1$

50 is closer to 50.41 (0.41 away) than to 49 (1 away).

Example 3: Estimate $\sqrt[3]{30}$ to 1 decimal place

1. **Trap:** $27 < 30 < 64$

Perfect *cubes* either side: $3^3 = 27$, $4^3 = 64$.

2. **Roots:** $3 < \sqrt[3]{30} < 4$

Take the cube root of each.

3. **Closer?** closer to 27

30 is 3 away from 27 but 34 away from 64, so $\sqrt[3]{30}$ is just over 3.

4. **Test:** $3.1^3 = 29.791$, $3.2^3 = 32.768$

30 is between these, so $\sqrt[3]{30}$ is between 3.1 and 3.2.

5. **Decide:** $\sqrt[3]{30} \approx 3.1$

30 is much closer to 29.791 than to 32.768.

My notes:

Support Box

Calculating a root

1. Square root: ask “what number $\times$ itself = n ?”
2. Cube root: ask “what number $\times$ itself $\times$ itself = n ?”
3. Check by squaring or cubing your answer.
4. $\sqrt{\text{negative}}$: no real answer. $\sqrt[3]{\text{negative}}$: a negative answer.

Estimating a root

1. **Trap** n between two perfect squares (or cubes).
2. **Roots**: write the two whole numbers.
3. **Closer?** Which perfect square/cube is n nearer?
4. **Test** decimals by squaring/cubing.
5. **Decide** and write $\approx$.

Sentence stem: “ $\sqrt{\quad}$ is between $\underline{\quad}$ and $\underline{\quad}$ because $\underline{\quad} < \underline{\quad} < \underline{\quad}$.”

Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225 **Cubes:** 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000

Calculator: you may use it to multiply (e.g. 4.5×4.5) but *not* the $\sqrt{\quad}$ or $\sqrt[3]{\quad}$ keys, until your teacher says “check”.

Section 4: WE DO

together · support fades

W1. (*Fill in the blanks.*)

(a) $\sqrt{144} = \underline{\quad}$ because $\underline{\quad}^2 = 144$

(b) $\sqrt[3]{64} = \underline{\quad}$ because $\underline{\quad}^3 = 64$

(c) $\sqrt[3]{-27} = \underline{\quad}$ because $(\underline{\quad})^3 = -27$

W2. Estimate $\sqrt{20}$ to 1 decimal place. (*Some steps are started for you.*)

Trap: $\underline{\quad} < 20 < \underline{\quad}$ **Roots:** $\underline{\quad} < \sqrt{20} < \underline{\quad}$ **Closer to:** $\underline{\quad}$

Test: $4.4^2 = \underline{\quad}$ $4.5^2 = \underline{\quad}$ **Decide:** $\sqrt{20} \approx \underline{\quad}$

W3. Estimate $\sqrt[3]{100}$ to 1 decimal place. (*Use the five steps: Trap, Roots, Closer?, Test, Decide.*)

Section 5: YOU DO*independent · about 10 minutes*

Must: A1–A8 and B1–B4. **Should:** B5–B8. **Could:** Section C. **Extension:** Section D (optional).

A. Foundation

A1. $\sqrt{25} = \underline{\hspace{2cm}}$ **A2.** $\sqrt{100} = \underline{\hspace{2cm}}$ **A3.** $\sqrt{121} = \underline{\hspace{2cm}}$ **A4.** $\sqrt[3]{8} = \underline{\hspace{2cm}}$

A5. $\sqrt[3]{1000} = \underline{\hspace{2cm}}$ **A6.** $\sqrt[3]{-64} = \underline{\hspace{2cm}}$ **A7.** $\sqrt{169} = \underline{\hspace{2cm}}$

A8. Between which two consecutive whole numbers does $\sqrt{18}$ lie? $\underline{\hspace{2cm}} < \sqrt{18} < \underline{\hspace{2cm}}$ because $\underline{\hspace{2cm}} < 18 < \underline{\hspace{2cm}}$

B. Core

B1. $\sqrt{225} = \underline{\hspace{2cm}}$ **B2.** $\sqrt[3]{216} = \underline{\hspace{2cm}}$ **B3.** $\sqrt[3]{-125} = \underline{\hspace{2cm}}$

B4. Explain why $\sqrt{-16}$ is not a real number.

B5. Between which two consecutive whole numbers does $\sqrt[3]{40}$ lie? Show the trap.

B6. Estimate $\sqrt{10}$ to 1 d.p.

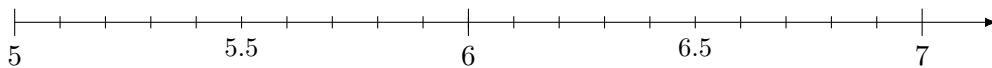
B7. Estimate $\sqrt{75}$ to 1 d.p.

B8. Estimate $\sqrt[3]{20}$ to 1 d.p.

C. Apply and reason

- C1.** A square garden has area 196 m^2 . Find the length of one side.
- C2.** A cube-shaped box has volume 343 cm^3 . Find the length of one edge.

- C3.** Without using the $\sqrt{\quad}$ or $\sqrt[3]{\quad}$ keys, mark these on the number line and then write them in order from smallest to largest: $\sqrt{40}$, 6.5 , $\sqrt[3]{200}$, 6



Order: _____

- C4.** A square tile has area 60 cm^2 . Estimate its side length to 1 decimal place. Show a check.

- C5.** A cube has volume 64 cm^3 . A square has area 64 cm^2 . Which has the longer side or edge? Justify your answer.

D. Challenge (optional)

- D1.** Find (a) $\sqrt{\frac{9}{16}}$ (b) $\sqrt[3]{\frac{1}{8}}$ *Hint: take the root of the top and of the bottom, e.g. $\sqrt{\frac{4}{25}} = \frac{\sqrt{4}}{\sqrt{25}} = \frac{2}{5}$.*

- D2.** Maya says: “Taking a square root always makes a number smaller.” Test her claim with 0.25 and with 1. Is she right?

- D3.** Find a whole number n so that $\sqrt{n}$ is between 9.5 and 10. How many such whole numbers are there?

Section 6: Spot the Mistake*pairs, then whole class · 3 minutes*

For each one: (i) circle the **first** error, (ii) explain what went wrong, (iii) write the correct answer.

E1. Sam's work

$$\sqrt{36} = 18 \text{ because } 36 \div 2 = 18.$$

E2. Lina's work

$$\sqrt{9 + 16} = \sqrt{9} + \sqrt{16} = 3 + 4 = 7$$

E3. Omar's work: estimate $\sqrt{40}$ to 1 d.p.

$$36 < 40 < 49, \text{ so } 6 < \sqrt{40} < 7. \quad 40 \text{ is between } 36 \text{ and } 49, \text{ so } \sqrt{40} \text{ is halfway.} \quad \sqrt{40} \approx 6.5$$

Exit ticket on the next page. Your teacher will collect it.

Exit Ticket*independent · 4 minutes · no calculator*

Name: _____

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X1. Calculate (a) $\sqrt{49}$ (b) $\sqrt[3]{27}$ **X2.** Calculate $\sqrt[3]{-1000}$. Show a check.**X3.** (a) Between which two consecutive whole numbers does $\sqrt{85}$ lie? (b) Estimate $\sqrt{85}$ to 1 decimal place.**X4.** A cube has volume 90 cm^3 . Between which two whole numbers is the length of its edge? Explain.**How did it go? Circle one.**

I can calculate and estimate roots:

on my own

with the Support Box

I need more help